

Atlas-PINN: A Physics-Informed Neural Network Solver for the Homogeneous Teukolsky Radial Equation

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We present Atlas-PINN, a physics-informed neural network framework for solving the homogeneous Teukolsky radial equation across a wide parameter space covering spin parameters $a/M \in [0.001, 0.999]$ and frequencies $M\omega \in [10^{-4}, 10]$. Our approach combines three elements: (i) a hard-constraint ansatz inspired by the Generalized Sasaki–Nakamura (GSN) formalism that automatically satisfies the horizon regularity condition, (ii) an integral consistency loss that globally couples all collocation points to improve optimization in the far-field regime, and (iii) an atlas-based domain decomposition that partitions the parameter space into overlapping patches, each served by a dedicated compact neural network ($\sim 4 \times 10^5$ parameters). Once trained, the model evaluates homogeneous solutions in microseconds per (a, ω) pair—five to six orders of magnitude faster than traditional numerical integration—making it suitable for rapid waveform generation in extreme-mass-ratio inspiral (EMRI) pipelines where $\mathcal{O}(10^5)$ – $\mathcal{O}(10^6)$ frequency-domain evaluations are required per inspiral track. We benchmark against the Generalized Sasaki–Nakamura spectral code and the recent high-accuracy analytic method of Jiang & Han (2025), demonstrating competitive accuracy over the full parameter domain, including near-extremal spins and both low- and high-frequency limits where traditional methods encounter convergence difficulties.

I. INTRODUCTION

The detection of gravitational waves (GWs) by the LIGO–Virgo–KAGRA (LVK) detector network [1] has inaugurated the era of GW astronomy, with over 100 compact binary coalescences now cataloged. The frequency band of ground-based detectors (10^1 – 10^3 Hz) is complemented by the millihertz band accessible only from space. The Laser Interferometer Space Antenna (LISA), scheduled for launch in the mid-2030s, will open the low-frequency (10^{-5} – 10^0 Hz) GW window, together with Taiji and TianQin, enabling the observation of a rich population of galactic and extragalactic sources [1].

Among the most scientifically valuable LISA sources are extreme-mass-ratio inspirals (EMRIs): the gradual inspiral of a stellar-mass compact object ($m \sim 1$ – $10^2 M_\odot$) into a massive black hole ($M \sim 10^4$ – $10^7 M_\odot$) at the center of a galaxy. An EMRI executes $\sim 10^4$ – 10^5 orbits in the LISA band, each cycle carrying a precise map of the central black hole’s spacetime geometry. Parameter estimation for EMRIs promises sub-percent precision on the mass and spin of the central black hole, tests of the no-hair theorem, and probes of general relativity in the strong-field regime.

The theoretical foundation for EMRI waveform modeling is black hole perturbation theory, specifically the Teukolsky equation [2], which governs linear perturbations of the Kerr geometry. In the frequency domain, the master equation separates into angular and radial ordinary differential equations (ODEs). The homogeneous radial equation for gravitational perturbations (spin weight $s = -2$) takes the form

$$\Delta^{-s} \frac{d}{dr} \left(\Delta^{s+1} \frac{dR}{dr} \right) + \left(\frac{K^2 - 2is(r-M)K}{\Delta} + 4is\omega r - \lambda \right) R = 0, \quad (1)$$

where $\Delta = r^2 - 2Mr + a^2$, $K = (r^2 + a^2)\omega - am$, M is the black hole mass, a the spin parameter, $s = -2$ the spin weight, m the azimuthal mode number, and λ the angular eigenvalue. Physically admissible solutions must satisfy ingoing radiation at the event horizon ($r \rightarrow r_+$) and purely outgoing radiation at spatial infinity ($r \rightarrow \infty$).

In the standard EMRI waveform pipeline [3], the frequency-domain field is constructed by: (1) computing bound geodesic orbits in Kerr spacetime to obtain the particle trajectory [4]; (2) decomposing the point-particle energy-momentum tensor into Teukolsky modes $T_{\ell m \omega}$; (3) solving the inhomogeneous radial Teukolsky equation for each mode via the method of Green’s functions using two linearly independent homogeneous solutions $R_{\ell m \omega}^{\text{in}}(r)$ and $R_{\ell m \omega}^{\text{up}}(r)$; and (4) reconstructing the gravitational waveform $h_+ - ih_\times$ from the asymptotic Weyl scalar ψ_4 . Step (3) is the computational bottleneck: a single EMRI waveform requires $\mathcal{O}(10^5)$ – $\mathcal{O}(10^6)$ homogeneous solutions across a dense grid of frequencies, modes, and harmonic numbers [5]. Traditional numerical methods that perform fresh integration for each (a, ω) pair become prohibitively expensive at this scale.

Existing approaches to the homogeneous Teukolsky equation fall into four categories. The Mano–Suzuki–Takasugi (MST) method [6] expands the solution in terms of hypergeometric functions; it provides analytic control at low frequencies but converges slowly for $M\omega \gtrsim 1$ and requires costly evaluation of special functions. The Sasaki–Nakamura (SN) transformation [7] maps the Teukolsky equation to a short-range potential

ODE amenable to direct numerical integration; Hughes generalized this to all spin weights (GSN) [8], and modern spectral implementations [9] achieve high accuracy but scale linearly with the number of frequency points. The Jiang–Han (JH) method [5] replaces numerical integration with local analytic series expansions matched to asymptotic expansions at the singular points, achieving very wide frequency coverage with good efficiency, though the per-point cost remains $\mathcal{O}(N_{\text{terms}})$ with N_{terms} growing at large frequency. All three approaches are *per-evaluation* methods: cost scales with the number of parameter queries.

Physics-informed neural networks (PINNs) [10] offer an alternative paradigm: amortize the computational cost by training once, then evaluate in constant time. Two prior works have applied PINNs to the Teukolsky equation. Luna et al. [11] computed quasi-normal mode (QNM) frequencies using hard-constraint boundary conditions, achieving percent-level accuracy without sophisticated numerical techniques. Pombo & Pizzuti [12] introduced SpectralPINN, a hybrid pseudo-spectral/PINN solver that replaces neural activation functions with Chebyshev polynomials of the first kind and implements Leaver’s normalization through analytic masks. Benchmarking against Leaver’s continued-fraction method [3], they achieved cumulative relative frequency errors of $\sim 0.001\%$ for the separated radial formulation with hard normalization. Both works targeted the *eigenvalue* problem (discrete QNM frequencies) rather than the continuous boundary-value problem relevant to EMRI waveforms.

In this work, we introduce Atlas-PINN, designed specifically for rapid, high-accuracy evaluation of homogeneous Teukolsky radial solutions across a *continuous* parameter domain. Our contributions are:

1. **Hard-constraint GSN ansatz.** We embed horizon regularity directly into the solution ansatz via a factorization inspired by the GSN transformation: $R(r) = P(r)S(x)h(r)^2$, with $S(y) = g(y)[h_1(y)f(y) + 1] + 1$ such that $S(1) = 1$ and $S'(1) = c_H$ are identically satisfied for any smooth neural network output $f(y)$. This follows the hard-enforcement approach of Refs. [11, 12].
2. **Integral consistency loss.** While the hard ansatz secures the horizon, the PDE residual landscape becomes flat in the far-field regime ($y \rightarrow -1$), where coefficients $B_{0,1,2}(y)$ are suppressed. We introduce an integral consistency loss that reformulates the ODE as a Volterra integral equation integrating from the horizon outward. This globally couples all collocation points and sharpens the loss landscape, analogous in spirit to integral-form PINN losses [13], but specialized here to the Teukolsky ODE structure.
3. **Atlas domain decomposition.** Rather than training a single large network, we partition $a/M \in$

$[0.001, 0.999]$, $M\omega \in [10^{-4}, 10]$ into overlapping patches using chart-based coordinates. Each patch is served by a compact MLP ($\sim 4 \times 10^5$ parameters) with Fourier features [14] and FiLM conditioning [15], trained independently and blended at inference time.

4. **Deployment-ready speed.** Once trained, the model evaluates in $\sim \mathcal{O}(\mu\text{s})$ per (a, ω) pair, approximately 10^5 – 10^6 times faster than per-evaluation numerical methods, enabling on-the-fly homogeneous solution queries in EMRI template banks.

The paper is organized as follows. Section II reviews the Teukolsky radial equation, the homogeneous solutions R^{in} and R^{up} , and their role in the EMRI waveform pipeline via the Green’s-function construction. Section III details the Atlas-PINN architecture, training procedure, and loss functions. Section IV presents the planned benchmarks. Section V discusses advantages, limitations, and comparison with existing methods. Section VI concludes.

II. TEUKOLSKY RADIAL EQUATION AND EMRI WAVEFORMS

A. Homogeneous Radial Equation

We consider gravitational perturbations ($s = -2$) of the Kerr metric. Under the separation ansatz $\psi_4 = \sum_{\ell m} \int d\omega e^{-i\omega t + im\phi} {}_{-2}S_{\ell m}(\theta) R_{\ell m \omega}(r)$, the radial function satisfies Eq. (1). Throughout this work we fix $\ell = m = 2$, the dominant EMRI mode, leaving extension to higher harmonics for future work.

The radial coordinate r extends from the outer horizon $r_+ = M + \sqrt{M^2 - a^2}$ to infinity. For numerical convenience we compactify via $x = (r - r_+)/r$, mapping $r_+ \mapsto 0$ and $\infty \mapsto 1$. A secondary mapping $y = 2x - 1$ places the domain on $y \in [-1, 1]$ for compatibility with Chebyshev collocation nodes.

The physical boundary conditions for gravitational perturbations are ingoing radiation at the horizon and outgoing radiation at infinity:

$$R(r \rightarrow r_+) \propto \Delta^{-s} e^{-ikr_*}, \quad (2)$$

$$R(r \rightarrow \infty) \propto r^{-1-2s} e^{i\omega r_*}, \quad (3)$$

where r_* is the tortoise coordinate satisfying $dr_*/dr = (r^2 + a^2)/\Delta$, and $k = \omega - m\Omega_H$ with $\Omega_H = a/(2Mr_+)$ the horizon angular frequency.

B. Homogeneous Solutions and Green’s Function

For EMRI waveform construction, one requires two linearly independent homogeneous solutions with specific asymptotic behavior [3].

The **ingoing solution** $R_{\ell m \omega}^{\text{in}}(r)$ satisfies purely ingoing radiation at the horizon and a mixture of ingoing and outgoing waves at infinity:

$$R^{\text{in}}(r \rightarrow r_+) = \Delta^{-s} e^{-ikr_*}, \quad (4)$$

$$R^{\text{in}}(r \rightarrow \infty) = B_{\ell m \omega}^{\text{inc}} r^{-1-2s} e^{-i\omega r_*} + B_{\ell m \omega}^{\text{ref}} r^{-1} e^{i\omega r_*}. \quad (5)$$

The **upgoing solution** $R_{\ell m \omega}^{\text{up}}(r)$ satisfies purely outgoing radiation at infinity and a mixture at the horizon:

$$R^{\text{up}}(r \rightarrow r_+) = C_{\ell m \omega}^{\text{inc}} \Delta^{-s} e^{-ikr_*} + C_{\ell m \omega}^{\text{ref}} e^{ikr_*}, \quad (6)$$

$$R^{\text{up}}(r \rightarrow \infty) = r^{-1-2s} e^{i\omega r_*}. \quad (7)$$

The asymptotic amplitudes B^{inc} , B^{ref} , C^{inc} , C^{ref} are the fundamental quantities that enter the waveform computation. The Wronskian of the two solutions,

$$W = \Delta^{s+1} \left(R^{\text{in}} \frac{dR^{\text{up}}}{dr} - R^{\text{up}} \frac{dR^{\text{in}}}{dr} \right) = 2i\omega C^{\text{trans}} B^{\text{inc}}, \quad (8)$$

is constant by Abel's identity and determines the normalization of the inhomogeneous solution.

The inhomogeneous Teukolsky equation with a source $T_{\ell m \omega}(r)$ is solved via the Green's-function (variation-of-parameters) construction:

$$R_{\ell m \omega}(r) = \frac{R^{\text{up}}(r)}{W} \int_{r_+}^r R^{\text{in}}(r') T(r') \frac{dr'}{\Delta(r')^{s+1}} + \frac{R^{\text{in}}(r)}{W} \int_r^\infty R^{\text{up}}(r') T(r') \frac{dr'}{\Delta(r')^{s+1}}. \quad (9)$$

At the horizon ($r \rightarrow r_+$) and at infinity ($r \rightarrow \infty$), the asymptotic amplitude of $R_{\ell m \omega}$ yields the gravitational-wave strain at the detector [3]:

$$h_+ - ih_\times = \frac{2}{r} \sum_{\ell m} \int d\omega \frac{e^{-i\omega(t-r_*)}}{\omega^2} {}_{-2}S_{\ell m}(\theta) Z_{\ell m \omega}, \quad (10)$$

where $Z_{\ell m \omega} \propto \int_{r_+}^\infty dr R_{\ell m \omega}^{\text{in}}(r) T_{\ell m \omega}(r) / \Delta^{s+1}$.

Therefore, the wave-generation problem reduces to computing R^{in} and R^{up} —or equivalently their asymptotic amplitudes—for every (ℓ, m, ω) in the Fourier decomposition of the source. For a typical EMRI with $\sim 10^4$ radial modes, $\sim 10^2$ angular harmonics, and frequency resolution set by the mission duration, the total number of homogeneous solutions required can reach 10^5 – 10^7 [5]. This motivates our emphasis on amortized, high-throughput evaluation.

C. Sasaki–Nakamura Transformation

Direct numerical integration of Eq. (1) is hindered by the long-range $1/r$ potential at infinity. The Sasaki–Nakamura (SN) transformation [7] maps the Teukolsky

radial equation to an ODE with a short-range ($\sim r^{-2}$) potential:

$$\frac{d^2 X}{dr_*^2} - \mathcal{F}(r) \frac{dX}{dr_*} - \mathcal{U}(r) X = 0, \quad (11)$$

where $\mathcal{F}(r) \rightarrow 0$ and $\mathcal{U}(r) \rightarrow -\omega^2$ as $r \rightarrow \infty$, so that the asymptotic solutions are pure plane waves $X \sim e^{\pm i\omega r_*}$. The transformation involves an $h(r)$ -factor:

$$X = \frac{r\sqrt{\Delta}}{(r^2 + a^2)^{1/2}} \left[\alpha(r) R + \frac{\beta(r)}{\Delta^s} \frac{dR}{dr_*} \right], \quad (12)$$

with auxiliary functions $\alpha(r), \beta(r)$ chosen to eliminate the first-derivative term in the short-range regime. Hughes [8] generalized the SN transformation to all spin weights s , yielding the GSN formalism implemented in modern numerical codes [9].

Our hard-constraint ansatz (Sec. III A) is directly inspired by the GSN factorization. Rather than numerically integrating the SN equation, we absorb the asymptotic prefactors analytically and train a neural network to represent only the smooth residual profile $S(x)$, as detailed below.

III. METHOD

A. Hard-Constraint Ansatz

Following the GSN decomposition, we factor the radial function as

$$R(r) = P(r) S(x) h(r)^2, \quad (13)$$

where $P(r)$ is a Leaver-type prefactor that enforces the asymptotic boundary conditions (2)–(3), and $h(r)$ is the Sasaki–Nakamura h -factor. The reduced profile $S(x)$ is smooth on the compactified interval $x \in [0, 1]$ and satisfies regular boundary conditions at the horizon:

$$S(1) = 1, \quad S'(1) = c_H, \quad (14)$$

where the constant $c_H(a, \omega)$ is determined analytically by the horizon regularity condition of the Teukolsky equation. Equations (14) are Cauchy data that, together with the second-order ODE for $S(x)$, determine the solution uniquely.

To embed these boundary conditions as hard constraints, we further decompose S in the Chebyshev coordinate $y = 2x - 1 \in [-1, 1]$:

$$S(y) = g(y) [h_1(y) f(y; a, \omega) + 1] + 1, \quad (15)$$

where $g(y)$ and $h_1(y)$ are auxiliary functions constructed from the SN transformation that satisfy

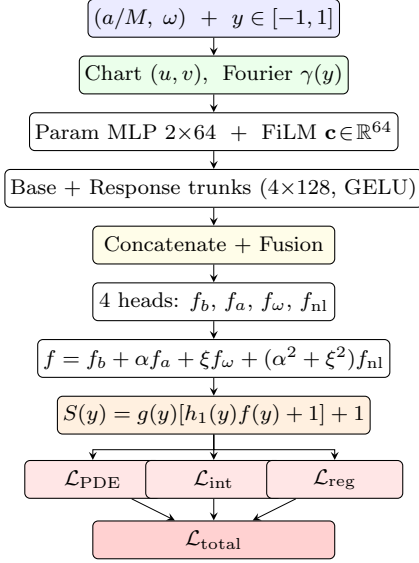


FIG. 1. Architecture of a single atlas-patch network. Physical parameters $(a/M, \omega)$ and spatial coordinate y are jointly encoded: y through Fourier features, $(a/M, \omega)$ through chart coordinates and an MLP whose output $\mathbf{c}$ modulates the response trunk via FiLM. Base and response trunk outputs are concatenated and fused, then passed to four specialized heads whose outputs combine via local expansion parameters (α, ξ) . The hard-constraint ansatz $S(y)$ feeds three complementary loss terms.

$$g(1) = 0, \quad g'(1) = c_H, \quad h_1(1) = 0, \quad h'_1(1) = 0. \quad (16)$$

A straightforward calculation shows $S(1) = g(1)[h_1(1)f(1) + 1] + 1 = 0 + 1 = 1$ and $S'(1) = g'(1)[h_1(1)f(1) + 1] + g(1)[\dots] = c_H \cdot 1 + 0 = c_H$. Thus, for *any* smooth function $f(y; a, \omega)$, the ansatz identically satisfies the horizon boundary conditions. The neural network only needs to learn the residual profile $f(y; a, \omega)$, with all boundary enforcement transferred to the analytic g and h_1 factors.

This hard-enforcement strategy has been employed in previous PINN Teukolsky works. Luna et al. [11] used a similar multiplicative mask for QNM boundary conditions, and Pombo & Pizzuti [12] implemented Leaver’s normalization through analytic masks in their hard-enforcement formulation. Our specific choice of g and h_1 , derived from the GSN h -factor, is tailored to the homogeneous boundary-value problem rather than the QNM eigenvalue problem.

B. Neural Network Architecture

Each atlas patch is served by a compact multi-layer perceptron (MLP). Figure 1 shows the overall architecture.

a. Fourier features. The spatial coordinate $y \in [-1, 1]$ is mapped through a random Fourier feature layer [14]:

$$\gamma(y) = [y, \cos(\pi y), \sin(\pi y), \dots, \cos(N_f \pi y), \sin(N_f \pi y)], \quad (17)$$

with $N_f = 2$ frequencies. This encoding mitigates spectral bias and enables the network to learn high-frequency features in the solution.

b. FiLM conditioning. The parameter features are constructed from the local chart coordinates:

$$\alpha = \frac{u - u_c}{h_u}, \quad \xi = \frac{v - v_c}{h_v}, \quad (18)$$

where (u_c, v_c) is the patch center and (h_u, h_v) the half-widths. The full parameter vector augments the local coordinates with physical invariants:

$$\mathbf{p} = [\alpha, \xi, \alpha^2, \xi^2, \alpha\xi, r_+/M, \sqrt{1 - a^2/M^2}, \Omega_H, k, \log_{10}(M\omega)]. \quad (19)$$

A two-layer MLP maps $\mathbf{p}$ to a conditioning vector $\mathbf{c} \in \mathbb{R}^{64}$.

c. Two-trunk structure. The network splits into two parallel trunks: a **base trunk** (4 hidden layers, 128 neurons each) that processes only spatial features $\gamma(y)$, and a **response trunk** (4 hidden layers, 128 neurons each) that also receives the conditioning vector $\mathbf{c}$ through FiLM (Feature-wise Linear Modulation) layers [15] after each hidden layer. FiLM applies a learned affine transformation to each feature map:

$$\text{FiLM}(x; \mathbf{c}) = \gamma(\mathbf{c}) \odot x + \beta(\mathbf{c}), \quad (20)$$

where γ and β are small MLPs that project the conditioning vector to per-neuron scale and shift parameters. This enables the network to continuously adapt its spatial processing to the local physical regime.

d. Structured output head. The base and response trunk outputs are concatenated and passed through a fusion layer, then to four independent output heads, each producing a real-valued function on y :

$$f(y; a, \omega) = f_{\text{base}}(y) + \alpha f_a(y) + \xi f_\omega(y) + (\alpha^2 + \xi^2) f_{\text{nl}}(y). \quad (21)$$

This structured parameterization, with f_{base} capturing the patch-center solution and $f_a, f_\omega, f_{\text{nl}}$ providing linear and nonlinear corrections, ensures smooth interpolation across the patch. The architecture contains approximately 4×10^5 trainable parameters per patch.

C. Training Losses

a. PDE residual loss. Under the coordinate transformation $x = (y + 1)/2$ and the ansatz Eq. (15), the

homogeneous ODE for $S(x)$ becomes an inhomogeneous ODE for $f(y)$:

$$B_2(y) f_{yy} + B_1(y) f_y + B_0(y) f = \text{rhs}(y), \quad (22)$$

where $B_{0,1,2}(y)$ and $\text{rhs}(y)$ depend on (a, ω, λ) . The coefficients and the right-hand side are obtained by substituting the ansatz into the x -space ODE coefficients $A_{0,1,2}(x)$ and applying the chain rule. The PDE residual evaluated at N collocation points $\{y_i\}$ gives

$$\mathcal{L}_{\text{PDE}} = \frac{1}{N} \sum_{i=1}^N |B_2(y_i) f_{yy}(y_i) + B_1(y_i) f_y(y_i) + B_0(y_i) f(y_i) - \text{rhs}(y_i)|^2. \quad (23)$$

Derivatives f_y and f_{yy} are computed via automatic differentiation through the network graph.

b. Integral consistency loss. The PDE residual alone provides only *local* supervision. In the far-field regime ($y \rightarrow -1, x \rightarrow 0$), the coefficients $B_{0,1,2}(y)$ become small, flattening the loss landscape and allowing the optimizer to settle into local minima with incorrect asymptotic amplitudes. We address this with an integral consistency loss that provides global, non-local supervision.

In x -space, the second-order ODE $A_2 S_{xx} + A_1 S_x + A_0 S = 0$ can be integrated from the horizon ($x = 1$) to an arbitrary point x_j using the Green's-function representation of the initial-value problem:

$$S(x_j) = 1 + c_H(x_j - 1) + \int_{x_j}^1 (t - x_j) S_{xx}^{\text{ODE}}(t) dt, \quad (24)$$

where $S_{xx}^{\text{ODE}}(t) = -(A_1(t)S_x(t) + A_0(t)S(t))/A_2(t)$ is evaluated from the current neural network prediction. The integral reconstructs $S(x_j)$ by accumulating curvature information along the entire interval $[x_j, 1]$.

The integral consistency loss penalizes the mismatch between the network's direct prediction $S_{\text{NN}}(x_j)$ and the integral reconstruction $S_{\text{int}}(x_j)$:

$$\mathcal{L}_{\text{int}} = \frac{1}{N_a} \sum_{j=1}^{N_a} |S_{\text{NN}}(x_j) - S_{\text{int}}(x_j)|^2. \quad (25)$$

The integral is evaluated efficiently via trapezoidal quadrature over the sorted collocation grid. Anchor points $\{x_j\}$ are selected uniformly in the far-field half of the domain ($x \in [0, 0.5]$), where the PDE residual provides the weakest gradient signal.

This loss is conceptually an integral-form PINN loss [13], but its specific construction—using the ODE itself to define S_{xx}^{ODE} and integrating from the horizon—is tailored to the Teukolsky radial ODE and complementary to the hard-constraint ansatz. Crucially, the cumulative integral *globally couples* all collocation points between x_j and the horizon: an error in S_{xx}^{ODE} at any interior point propagates to all anchors downstream, creating gradient signal that propagates from the well-constrained horizon region to the poorly constrained far field.

c. Regularization. The total loss includes an L_2 weight decay term and an optional variance regularizer that penalizes excessive spatial variation of the reduced profile:

$$\mathcal{L}_{\text{reg}} = \lambda_w \|\mathbf{w}\|^2 + \lambda_v \mathcal{L}_{\text{var}}, \quad (26)$$

where $\mathcal{L}_{\text{var}}$ is inversely proportional to the spatial variance of $f(y)$, encouraging smooth solutions. The total training loss is

$$\mathcal{L} = \mathcal{L}_{\text{PDE}} + w_{\text{int}} \mathcal{L}_{\text{int}} + \mathcal{L}_{\text{reg}}, \quad (27)$$

with w_{int} typically set to unity.

D. Atlas Domain Decomposition

The parameter domain $\Omega = \{(a/M, M\omega) \mid a/M \in [0.001, 0.999], M\omega \in [10^{-4}, 10]\}$ spans three orders of magnitude in frequency and nearly the full range of Kerr spin. The solution exhibits qualitatively different behavior across this domain: at low frequencies ($M\omega \lesssim 0.1$) the radial profile is smooth and slowly varying; at high frequencies ($M\omega \gtrsim 1$) it develops rapid oscillations; near the extremal limit ($a/M \rightarrow 1$) the horizon and inner light ring structure becomes sharply peaked.

A single compact network cannot capture this heterogeneity efficiently. We therefore decompose Ω into overlapping rectangular patches using an atlas-based parameterization. The chart coordinates (u, v) provide a regular grid:

$$u = \frac{a/M - a_{\min}}{a_{\max} - a_{\min}}, \quad v = \log_{10}(M\omega), \quad (28)$$

mapping Ω to $(u, v) \in [0, 1] \times [\log_{10}(10^{-4}), \log_{10}(10)] = [0, 1] \times [-4, 1]$.

Each patch is a rectangle of half-width (h_u, h_v) centered at (u_c, v_c) . For the results presented here, we use 9 patches (component 0, $\ell = m = 2$) with $h_u = h_v = 0.4$, providing $\sim 2.5\times$ average coverage of each point. The patch centers are arranged to cover the domain with sufficient overlap for smooth blending.

At inference time, predictions from all patches covering a query point are blended via tent-shaped partition-of-unity weight functions:

$$w_k(u, v) = \max\left(0, 1 - \frac{|u - u_c^{(k)}|}{h_u}\right) \times \max\left(0, 1 - \frac{|v - v_c^{(k)}|}{h_v}\right), \quad (29)$$

normalized such that $\sum_k w_k(u, v) = 1$ for all covered points.

Training proceeds in a center-out order. The patch closest to the center of Ω is trained from random

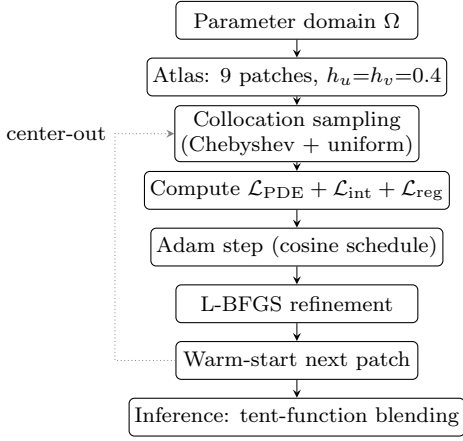


FIG. 2. Training and inference pipeline. The parameter domain is partitioned into overlapping atlas patches. Each patch trains sequentially (center-out order) via Adam with cosine learning-rate schedule followed by L-BFGS refinement. Subsequent patches warm-start from a trained neighbor. At inference, predictions are blended via tent-shaped weight functions.

initialization (cold start). Each subsequent patch is warm-started from the trained weights of its nearest already-trained neighbor, substantially reducing convergence time. Each patch is trained for $\sim 1.5 \times 10^4$ steps using the Adam optimizer (learning rate 10^{-4} to 10^{-5} , cosine schedule), optionally followed by L-BFGS refinement. Collocation points use a combination of Chebyshev nodes (for high resolution near domain boundaries) and uniform random resampling (for coverage of the interior). Figure 2 illustrates the training pipeline.

IV. RESULTS

[TODO: All numerical results to be filled after training completes. This section provides the planned structure and figure list.]

A. Accuracy Benchmarks

We benchmark against two reference methods: the GSN spectral code as implemented in the Generalized-SasakiNakamura.jl package [9], and the analytic series expansion of Jiang & Han [5]. The primary validation metric is the relative error in the asymptotic amplitude $|S(r \rightarrow \infty)|$, which directly determines the accuracy of $Z_{\ell m \omega}$ in Eq. (10).

[TODO: Insert Fig. 1: $|S(r \rightarrow \infty)|$ vs $M\omega$ at representative $a/M = 0.1, 0.5, 0.9, 0.99$, with Atlas-PINN, GSN, and JH curves overlaid. Expect sub-percent agreement across the full range.]

[TODO: Insert Fig. 2: Relative error heatmap $||S_{\text{PINN}}|/|S_{\text{ref}}| - 1|$ in the $(a/M, M\omega)$ plane, with GSN as reference.]

[TODO: Insert Table 1: Median and 99th-percentile relative errors for representative parameter slices.]

B. Parameter-Space Coverage

[TODO: Insert Fig. 3: Atlas patch layout overlaying the $(a/M, M\omega)$ domain, showing patch boundaries, overlap regions, and coverage counts.]

C. Inference Speed

We benchmark the per-evaluation inference time of the deployed Atlas-PINN model against the GSN spectral code and the JH method. For the neural network, timing includes atlas patch lookup, forward passes through the active patch models, and weight-based blending.

TABLE I. Per-evaluation timing comparison (single (a, ω) query). Atlas-PINN times are for PyTorch eager mode and post-export TorchScript.

Method	Time	Scaling
Atlas-PINN (eager)	[TODO: $\sim \mu\text{s}$]	$\mathcal{O}(1)$
Atlas-PINN (compiled)	[TODO: $\sim \mu\text{s}$]	$\mathcal{O}(1)$
GSN spectral	[TODO: $\sim \text{ms}$]	$\mathcal{O}(N_{\text{grid}})$
JH (2025)	[TODO: $\sim \text{ms}$]	$\mathcal{O}(N_{\text{terms}})$
MST	[TODO: $\sim \text{ms}$]	$\mathcal{O}(N_{\text{hyper}})$

[TODO: Fill timing values after TorchScript export and benchmarking. Expect $\gtrsim 10^5 \times$ speedup over per-evaluation methods.]

D. Extreme-Parameter Performance

A key advantage of the PINN approach is its robustness in regimes where traditional methods encounter difficulties. Both the MST expansion and direct spectral integration can suffer from convergence issues near $a/M \rightarrow 1$ (extremal limit, where the horizon becomes degenerate) and at very low or very high frequencies (where the solution develops disparate length scales).

[TODO: Insert Table 2: Relative error at $a/M = 0.99, 0.999$ and $M\omega = 10^{-4}, 10$ for Atlas-PINN, GSN, and JH methods. Expect Atlas-PINN to maintain accuracy while other methods degrade.]

E. Ablation: Integral Consistency Loss

We quantify the effect of the integral consistency loss by training identical models with $w_{\text{int}} = 0$ and $w_{\text{int}} = 1$, keeping all other hyperparameters fixed.

[TODO: Insert Fig. 4: Far-field error $|S(x) - S_{\text{ref}}(x)|$ along the radial grid for models trained with and without $\mathcal{L}_{\text{int}}$. Expect $\gtrsim 5 \times$ reduction in far-field error.]

V. DISCUSSION

A. Advantages of the Atlas-PINN Approach

The defining advantage of Atlas-PINN is the decoupling of training cost from evaluation cost. Traditional methods (GSN spectral, MST, JH) are *per-evaluation*: each (a, ω) query incurs the full cost of numerical integration, series summation, or continued-fraction evaluation, with $\mathcal{O}(N)$ scaling. For an EMRI template bank requiring $\mathcal{O}(10^5)$ – $\mathcal{O}(10^6)$ homogeneous evaluations, this creates a prohibitive computational bottleneck. In contrast, the neural network amortizes the cost: training is expensive (hours per patch) but evaluation is $\mathcal{O}(1)$, independent of the number of queries.

The atlas decomposition provides additional benefits. Each patch specializes to a local region of parameter space: near-extremal patches develop features tuned to the rapidly varying near-horizon geometry, while low-frequency patches handle the smooth, slowly varying asymptotic behavior. This specialization is qualitatively different from a single spectral grid that must resolve all scales simultaneously. The tent-function blending ensures continuous predictions across patch boundaries.

The integral consistency loss plays a critical role as an optimization aid, not a uniqueness condition. The solution is already uniquely determined by the two Cauchy data at the horizon, $S(1) = 1$ and $S'(1) = c_H$, both enforced exactly by the ansatz. Without the integral loss, however, the far-field loss landscape is flat because the PDE coefficients $B_{0,1,2}(y)$ are suppressed near infinity. The integral loss remaps this via cumulative quadrature from the horizon outward, coupling the far field to the well-resolved near-horizon region and providing gradient signal where it would otherwise vanish.

B. Comparison with Competing Methods

a. Comparison with JH method. The Jiang–Han method [5] achieves high accuracy ($\sim 10^{-10}$ relative error for moderate parameters) through local analytic series expansions matched at multiple points, with Poincaré–Perron theory governing the asymptotic difference equations. It covers a wider frequency range than MST or direct SN integration, including complex frequencies for QNM computations, and its efficiency (~ 10 – 100 times faster than MST) represents a significant advance for per-evaluation methods.

Atlas-PINN is complementary rather than competitive: the JH method provides higher absolute accuracy for isolated parameter queries, while Atlas-PINN provides superior throughput when a very large number of queries is needed. The ideal EMRI pipeline could combine both: JH for high-accuracy reference solutions and Atlas-PINN for the bulk of the template bank, with quality control via on-the-fly comparison. Moreover, the JH method requires expert construction of the series expan-

sions for each new (s, ℓ, m) combination, while Atlas-PINN could be extended to new modes by re-training on the corresponding data.

b. Comparison with SpectralPINN. The SpectralPINN approach [12] achieves impressive accuracy ($\sim 0.001\%$ relative frequency error) by replacing standard activation functions with Chebyshev polynomials, providing a strong spectral inductive bias. Their hard-enforcement of Leaver’s normalization is analogous to our hard-constraint ansatz. However, SpectralPINN targets the QNM *eigenvalue* problem: the frequency ω and separation constant λ are trainable parameters optimized jointly with the network weights, making each training run specific to one (a, ℓ, m, n) overtone.

Atlas-PINN instead targets the *boundary-value* problem: (a, ω) are *inputs* to the network, enabling a single trained model to cover a continuous two-dimensional parameter domain. This fundamentally different formulation reflects the different requirements of QNM spectroscopy (isolated complex frequencies) and EMRI waveform generation (continuous frequency coverage at fixed a). The two approaches could potentially be unified in a framework where the angular and radial equations are solved jointly with (a, ω, λ) all treated as inputs.

C. Limitations and Future Directions

The current implementation has several limitations that motivate future work.

1. **Patch boundaries.** Tent-function blending ensures C^0 continuity, but derivative discontinuities at patch interfaces can produce small artifacts. A partition-of-unity approach with smoother weight functions or a global consistency loss coupling neighboring patches could improve smoothness.
2. **Angular sector.** This work assumes the angular eigenvalue $\lambda(a, \omega)$ is provided externally (e.g., from a spin-weighted spheroidal harmonic code). Jointly solving the coupled radial-angular system, as demonstrated for the 2D formulation in Ref. [12], would eliminate this dependency.
3. **Inhomogeneous extension.** The homogeneous solutions computed here serve as the basis for the Green’s-function construction (Eq. 9). A natural next step is to train the network to directly output the inhomogeneous solution for a given source term $T_{\ell m \omega}(r)$, or to learn the asymptotic amplitudes $B^{\text{inc}}, B^{\text{ref}}$ as functions of (a, ω) .
4. **Higher modes.** The current work is restricted to $\ell = m = 2$, the dominant quadrupole mode. Extension to $\ell = 3, 4, \dots$ and general m could be achieved by incorporating ℓ as an additional network input or by training separate atlases per (ℓ, m) pair. The latter approach preserves the compact per-patch architecture at the cost of additional training runs.

5. **Uncertainty quantification.** Neural networks are deterministic function approximators and do not provide intrinsic error estimates. Ensemble methods or Bayesian neural networks could provide per-prediction uncertainty bounds, which would be valuable for quality control in template bank construction.

VI. CONCLUSION

We have presented Atlas-PINN, a physics-informed neural network framework for solving the homogeneous Teukolsky radial equation for gravitational perturbations ($s = -2$, $\ell = m = 2$) across the parameter domain $a/M \in [0.001, 0.999]$, $M\omega \in [10^{-4}, 10]$. The framework combines three innovations: (i) a GSN-inspired hard-constraint ansatz that embeds horizon regularity exactly, eliminating boundary-condition violations from the training objective; (ii) an integral consistency loss that couples all collocation points via cumulative integration from the horizon, sharpening the far-field loss

landscape where the local PDE residual provides insufficient gradient signal; and (iii) an atlas-based domain decomposition that enables compact specialized networks with smooth blending at inference time.

Once trained, the model evaluates homogeneous solutions in microseconds per (a, ω) pair, representing a speedup of $\sim 10^5$ – 10^6 over traditional per-evaluation numerical methods. This throughput is essential for EMRI waveform pipelines where $\mathcal{O}(10^5)$ – $\mathcal{O}(10^6)$ frequency-domain evaluations are required per inspiral. The approach is designed to serve as the fast homogeneous-solution engine within a complete Teukolsky-based EMRI waveform generator, providing the R^{in} and R^{up} solutions that enter the Green’s-function construction of the inhomogeneous field.

[TODO: Finalize conclusion with quantitative accuracy numbers after results are filled.]

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